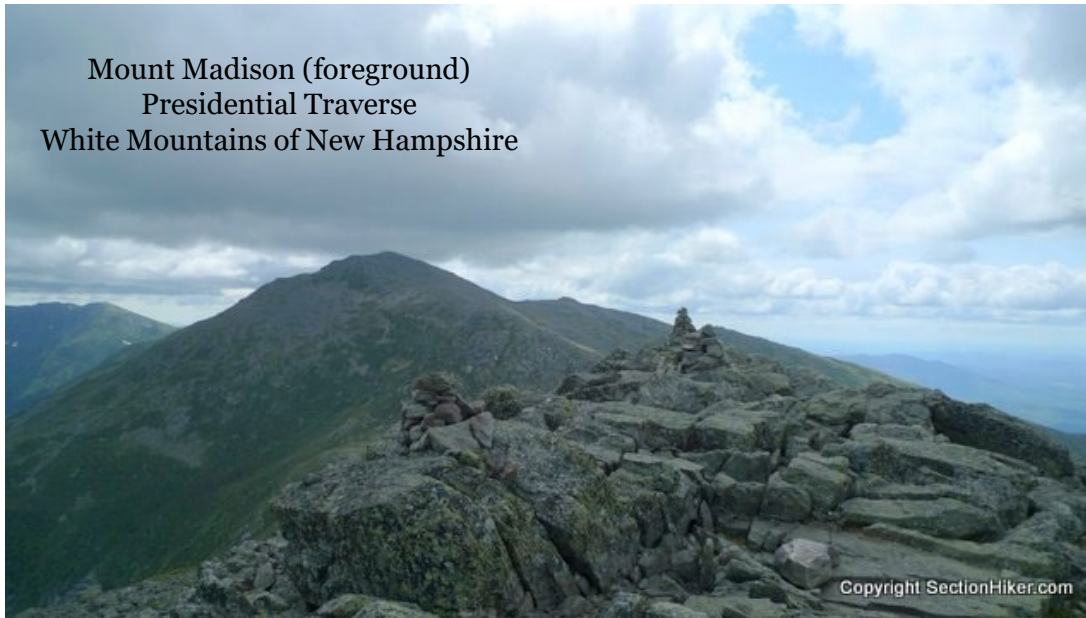


Multivariable Functions

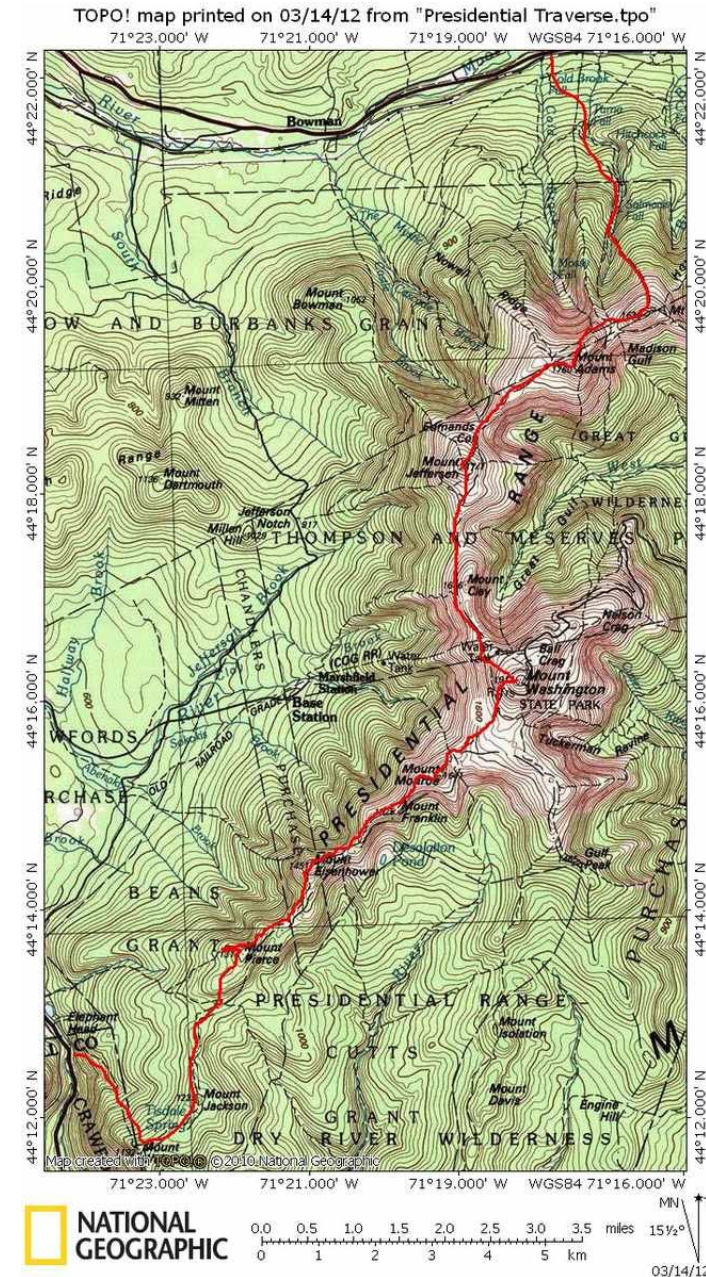
Trim Ch 11, S. 12.1

Mount Madison (foreground)
Presidential Traverse
White Mountains of New Hampshire



Copyright SectionHiker.com

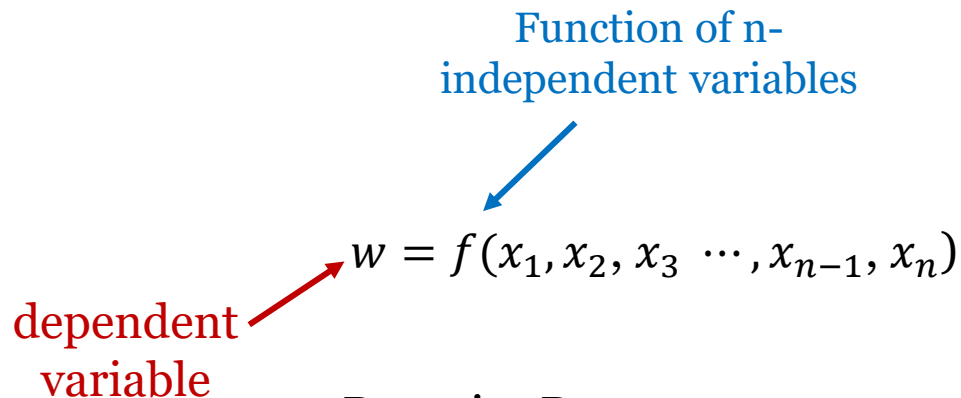
<http://sectionhiker.com/great-hikes-a-presidential-traverse/>



Multivariable Functions

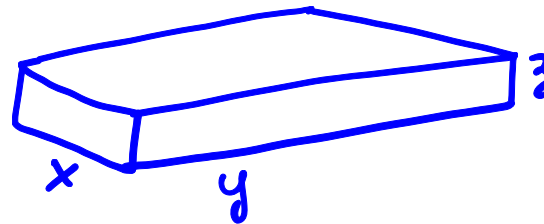
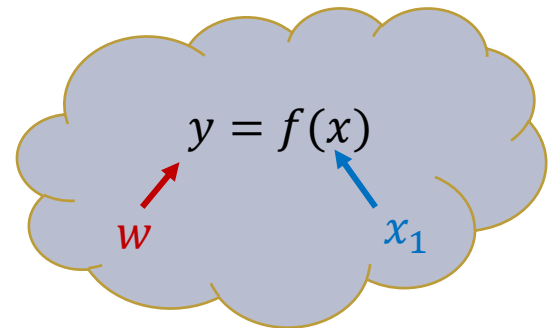
Suppose D is a set of n -tuples of real numbers $(x_1, x_2, x_3 \cdots, x_{n-1}, x_n)$

A **real valued function** f on D is a rule that assigns a unique (single) real number “ w ” to each element of D



Domain: D

Range: Set of all w



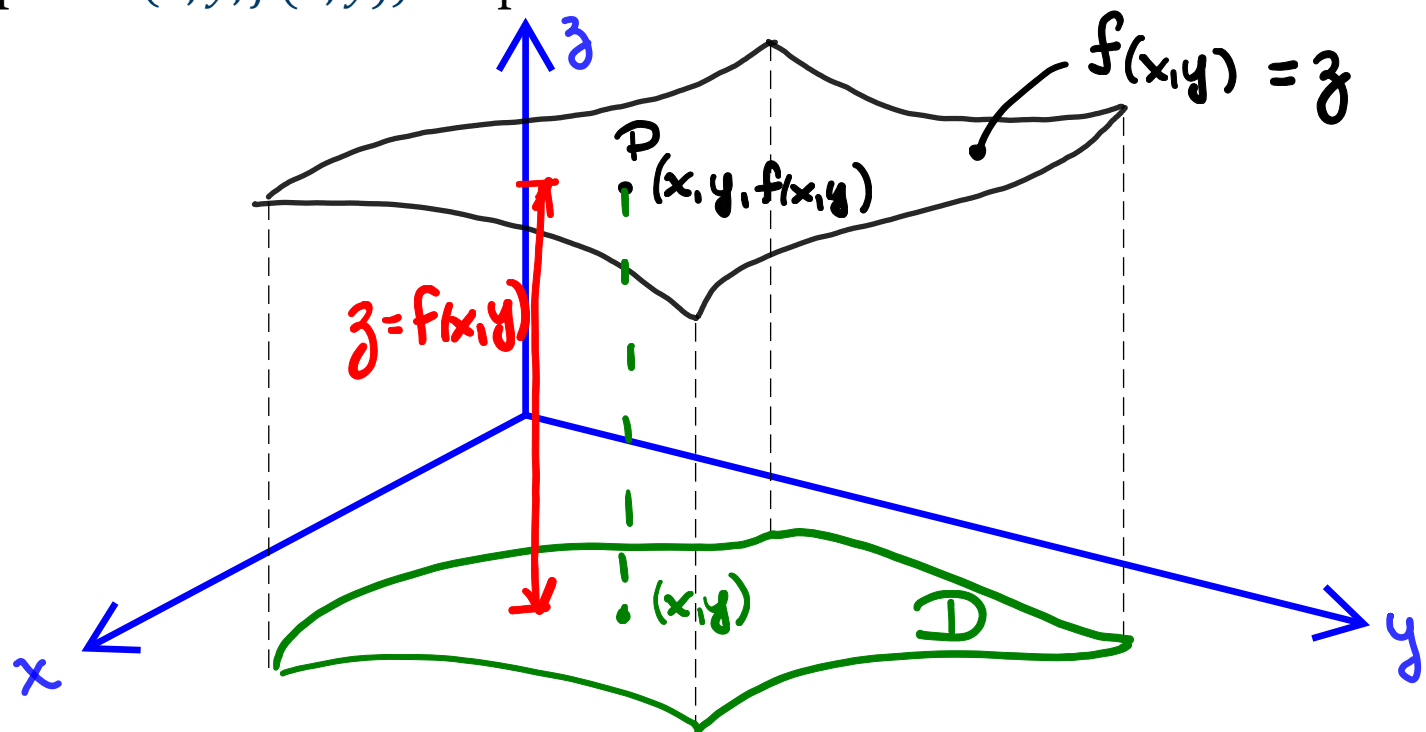
$$V(x, y, z) = xyz$$



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Graph of a Function of 2 Variables

A function of 2 variables $z = f(x, y)$ is the equation of a 3D surface formed by the set of all points $(x, y, f(x, y))$ in space.



1. The value of $z = f(x, y)$ is the height of the surface at $P(x, y)$
2. The domain of $f(x, y)$ is the set of all points in the $x - y$ plane that correspond to a unique value of $z = f(x, y)$



Contour and Level Curves of Functions of 2 Variables

There are two standard ways to visualize the values of a function $f(x, y)$. One is to sketch the surface $z = f(x, y)$ and another one is to draw and label curves in the domain on which $f(x, y)$ has a constant value.

Example:

Graph $f(x, y) = 100 - x^2 - y^2$ and plot the level and contour curves for $f(x, y) = 0$, $f(x, y) = 51$, $f(x, y) = 75$, in the domain of $f(x, y)$.

Solution:

DOMAIN : ENTIRE x - y PLANE

WRITING FUNCTION AS $z = f(x, y) = 100 - x^2 - y^2$

$$x^2 + y^2 - 100 = -z$$

$$z = 0 \rightarrow x^2 + y^2 = 100$$

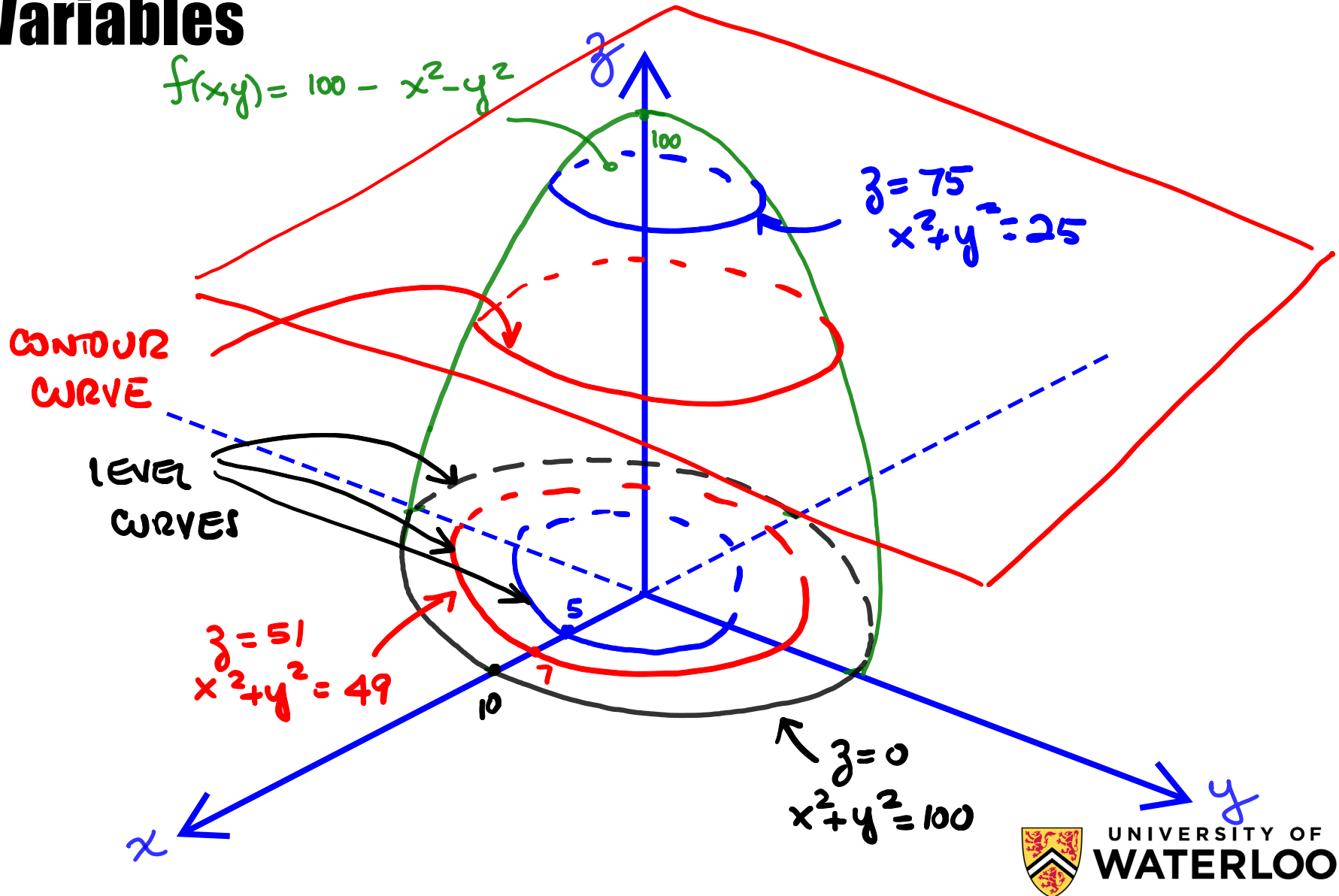
$$z = 51 \rightarrow x^2 + y^2 = 49$$

$$z = 75 \rightarrow x^2 + y^2 = 25$$



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Contour and Level Curves of Functions of 2 Variables



So far ...

$z = f(x, y)$ represents the height of a 3D surface at (x, y)

CONTOUR CURVE

A plane $z = c$ parallel to the $x - y$ plane and intersecting a surface $z = f(x, y)$ produces a *Contour Curve*.

LEVEL CURVE

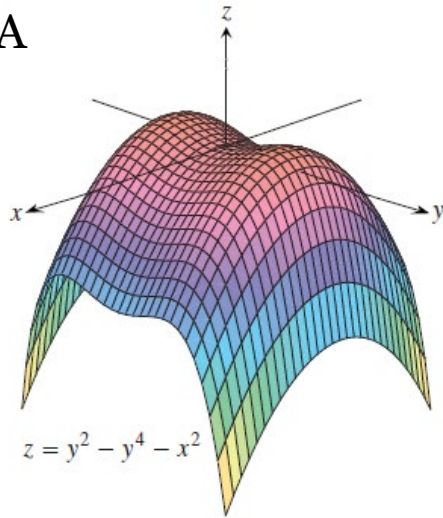
Set of points in the domain where a function $f(x, y)$ takes on a given value c .

CONTOUR MAP

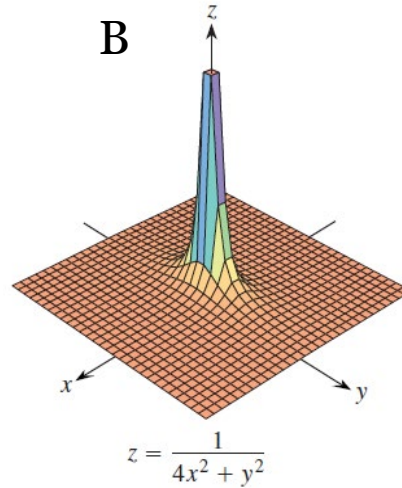
Plots where equidistant level curves are projected in the x - y plane, specifying their z -value

Match these functions of 2 variables

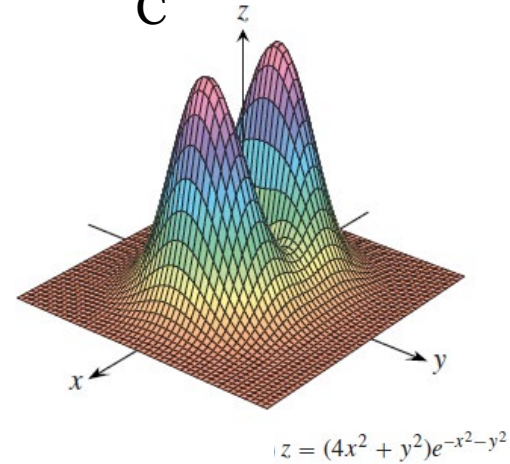
A



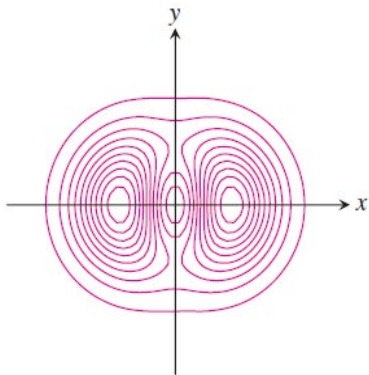
B



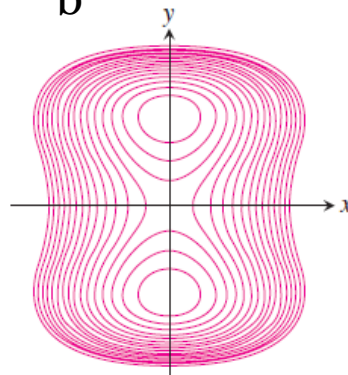
C



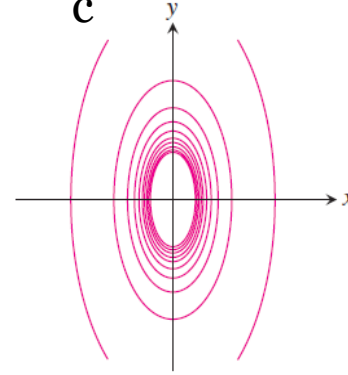
a



b



c



- a. Ab, Ba, Cc
- b. Ac, Bb, Ca
- c. Ab, Bc, Ca**
- d. Aa, Bb, Cc



Graphing Functions of 3 Variables

A function of three variables, f , is a rule that assigns to each ordered triple (x, y, z) in a domain D a unique real number denoted by $f(x, y, z)$.

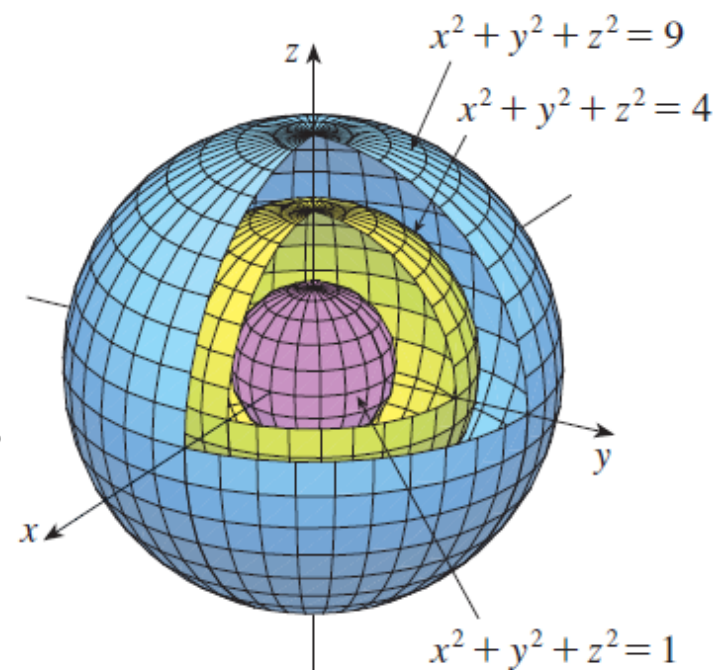
Example:

Graph the function:

$$f(x, y, z) = x^2 + y^2 + z^2$$

Solution:

The level surfaces are $x^2 + y^2 + z^2 = k$, where $k \geq 0$, and they form a family of concentric spheres with radius $\sqrt{k}$.



We are not graphing the function here; we are looking at level surfaces in the function's domain.



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Summary

- A **real valued function** f on D is a rule that assigns a unique (single) real number “ w ” to each element of D
- We use multivariable functions to **define real-life relationships**
- **Contour map** is a plot where equidistant level curves are projected in the $x - y$ plane. They are labeled by specifying their constant $z -$ value
- A function of three variables $f(x, y, z)$ lies in the 4D space and it can be “visualized” in 3D space by graphing their **level surfaces**



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Thank you